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**UNIVERSITÄT
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CAS ADS M1 Slides

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Data (latin datum in singular = thing given)

Data

Term first used in relation with computers in the 40-50ies

Plural but often used in singular

Means **any sequence of symbols** (or observable, measurable or statistically collectable values)

Needs processing and interpretation to become **information**

A lot of information and experience become **knowledge**

Digital (represented by 0 and 1) or analog

Moves serial or parallel

Stored on magnetic (tapes, hard disks), optical (CD, DVD) or mechanical devices

Most computers work on digital data and with the binary numeral system (“alphabet”)

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Data Examples

- Radius of the earth 46 100 km
- Survey responses
- Prices for same product in different shops
- Speech or music
- Natural Language
- Images
- Videos
- ...

Binary representation of “Hello”:

01001000 01100101 01101100 01101100 01101111 00100001



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Data Representations

Numeral Systems

Data is mostly represented by numeral systems

Writing systems for expressing numbers using e.g. digits

Modern systems are mostly positional systems

- $304 = 3 \times 10^2 + 0 \times 10^1 + 4 \times 10^0$

Examples Numeral Systems

Unary (2 = //) - base 1

- Good for kids

Binary (2=10) - base 2

- Good for computers

Decimal (2=2) - base 10

- Good for grown ups

Sexagecimal - base 60

- (remnants in hour, minutes etc)

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Most computers understand only 0/1

[A concept that] is not easy to impart to the pagans, is the creation *ex nihilo* through God's almighty power. Now one can say that nothing in the world can better present and demonstrate this power than the origin of numbers, as it is presented here through the simple and unadorned presentation of One and Zero or Nothing.

— Leibniz's letter to the [Duke of Brunswick](#) attached with the *I Ching* hexagrams^[19]

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Data

Binary numbers

Base 2, e.g. 0 and 1

Computers work with electrical currents, either there is a current (1) or there is no current (0)

Other numbers and characters can be represented with binary numbers

One can do mathematics with 0 and 1

So in the end data is mostly stored as bits and processed as bits

From 0 to 3

0000 0000 (8 bits = 1 byte)

0000 0001

0000 0010

0000 0011

....

1111 1111 ($2^8 = 256$)

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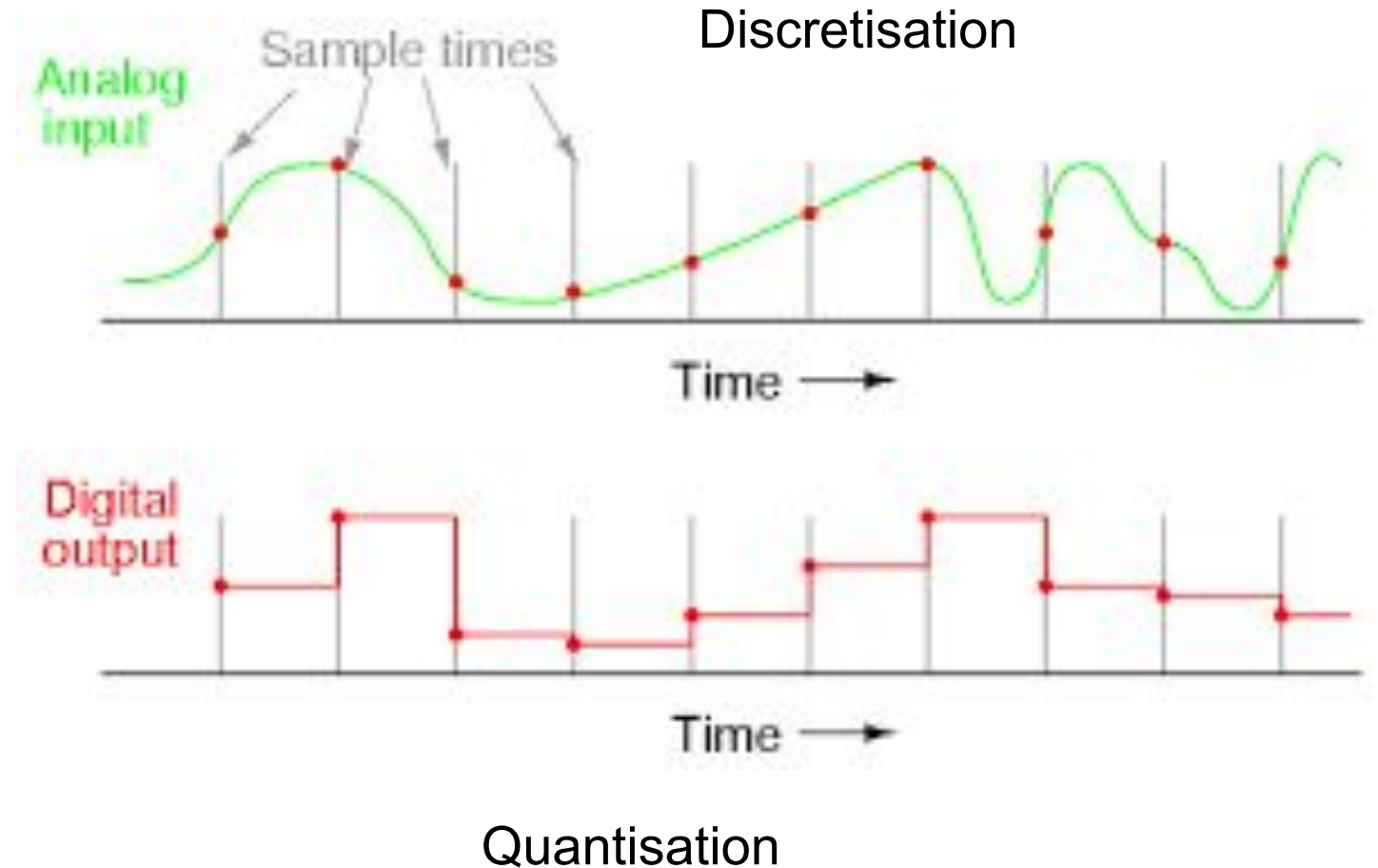
Analog to Digital Data (Digitisation) - ADC

Digitisation

Sensors often produce analog data - often perceived as some continuous wave

Is then often converted to digital data for computation and storage as bits

Lately an inflationary usage occurred - digitisation is now also a social process and every institution needs a digital strategy



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Data types and structures

Common types

Integer (e.g. 32-bit int)

Floating point (for real numbers), 32 or 64-bit (double) float

Boolean (True or False)

Character (a,b,c ...)

String ("list of characters")

List or Array ([1,2,r,t,5])

Explicit and implicit declaration

In programming languages data is loaded into variables of certain types

In Python and R types normally don't have to be specified.

- counter = 2

In C/C++, Fortran ... data types must be specified

- Int counter

- Float strength

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Data types and structures

Composite data types

Arrays, matrices, **dataframes**

In C/C++: Structures

- Person (name,age,gender)
- A structure with method (s) or functions are called classes
- An instance of a class is often called an object

Trees

Graphs

...

In R and in the Python pandas module, **dataframe** is the essential data (composite) type

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Data and some vocabulary

Prefixes for multiples of bits (bit) or bytes (B)				
Decimal		Binary		
Value	SI	Value	IEC	JEDEC
1000	k kilo	1024	Ki kibi	K kilo
1000 ²	M mega	1024 ²	Mi mebi	M mega
1000 ³	G giga	1024 ³	Gi gibi	G giga
1000 ⁴	T tera	1024 ⁴	Ti tebi	—
1000 ⁵	P peta	1024 ⁵	Pi pebi	—
1000 ⁶	E exa	1024 ⁶	Ei exbi	—
1000 ⁷	Z zetta	1024 ⁷	Zi zebi	—
1000 ⁸	Y yotta	1024 ⁸	Yi yobi	—

Unit prefixes

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Data volumes

Sizes

An integer number is often 4 bytes

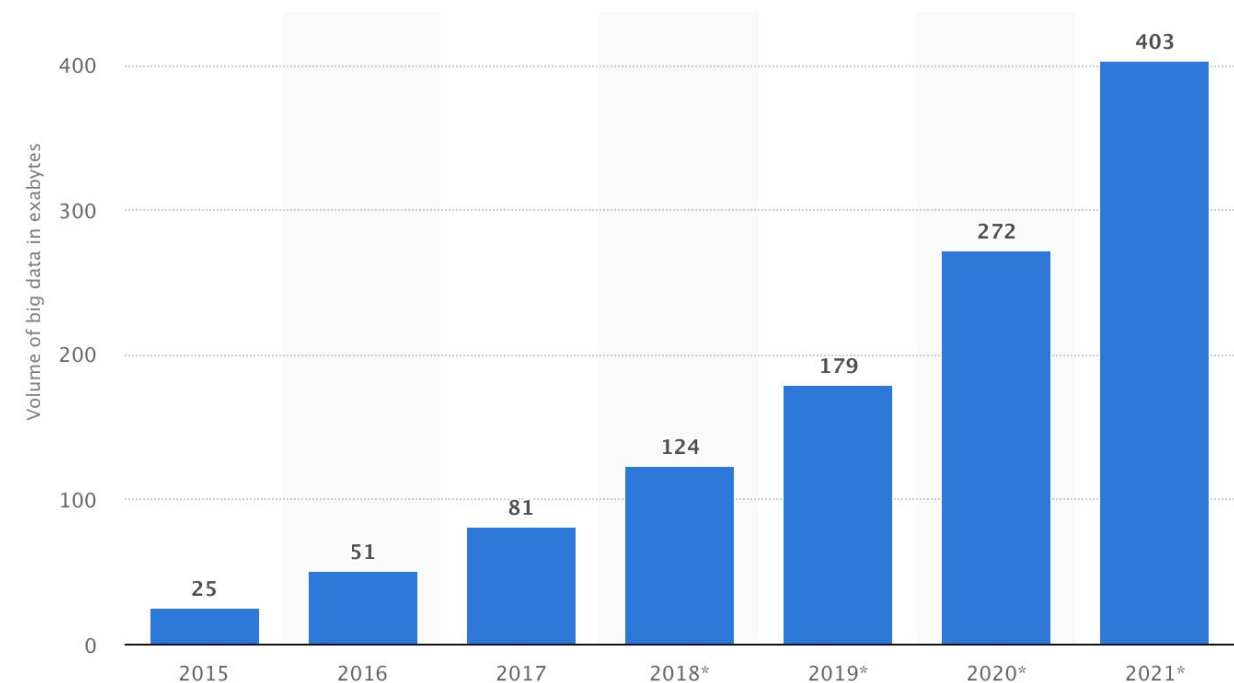
A character is often 1 byte

My 2017 laptop has 8GB RAM

In 2007 all digital data was 281 EB

In 10 years one expects 40 EB of new ge genomics data

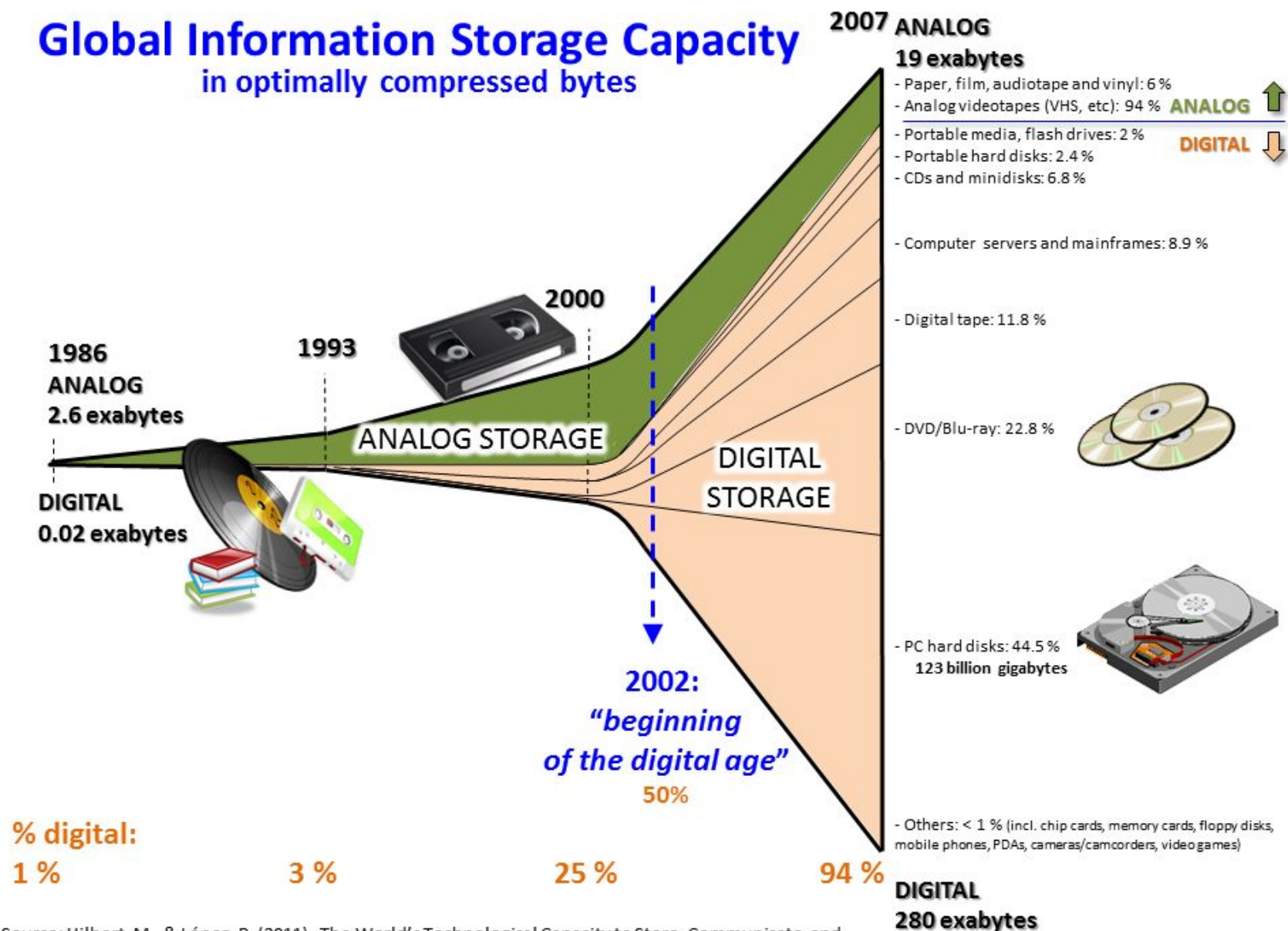
(Microscope image) - 3 numbers per pixel



- Statista.com : big data in data centers

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Global Information Storage Capacity in optimally compressed bytes



Source: Hilbert, M., & López, P. (2011). The World's Technological Capacity to Store, Communicate, and Compute Information. *Science*, 332(6025), 60 –65. <http://www.martinhilbert.net/WorldInfoCapacity.html>

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What is big data ?

Relative

Too big for your “traditional” tools

- Paper is not good for tens of rows
- Spreadsheets are not good for more than million rows
- Database management system may not be good for data more than the computer memory

Advanced processing

Parallel processing over distributed systems

Systems are orchestrated with HPC/SLURM, Hadoop, Spark ...

The processing is done with C++, Python, R or other programming languages with parallelism capabilities

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Metadata

Metadata is data about data

Value: 46 100

Unit: km (a good data scientist always provides the unit)

Author: Eratosthenes of Cyrene

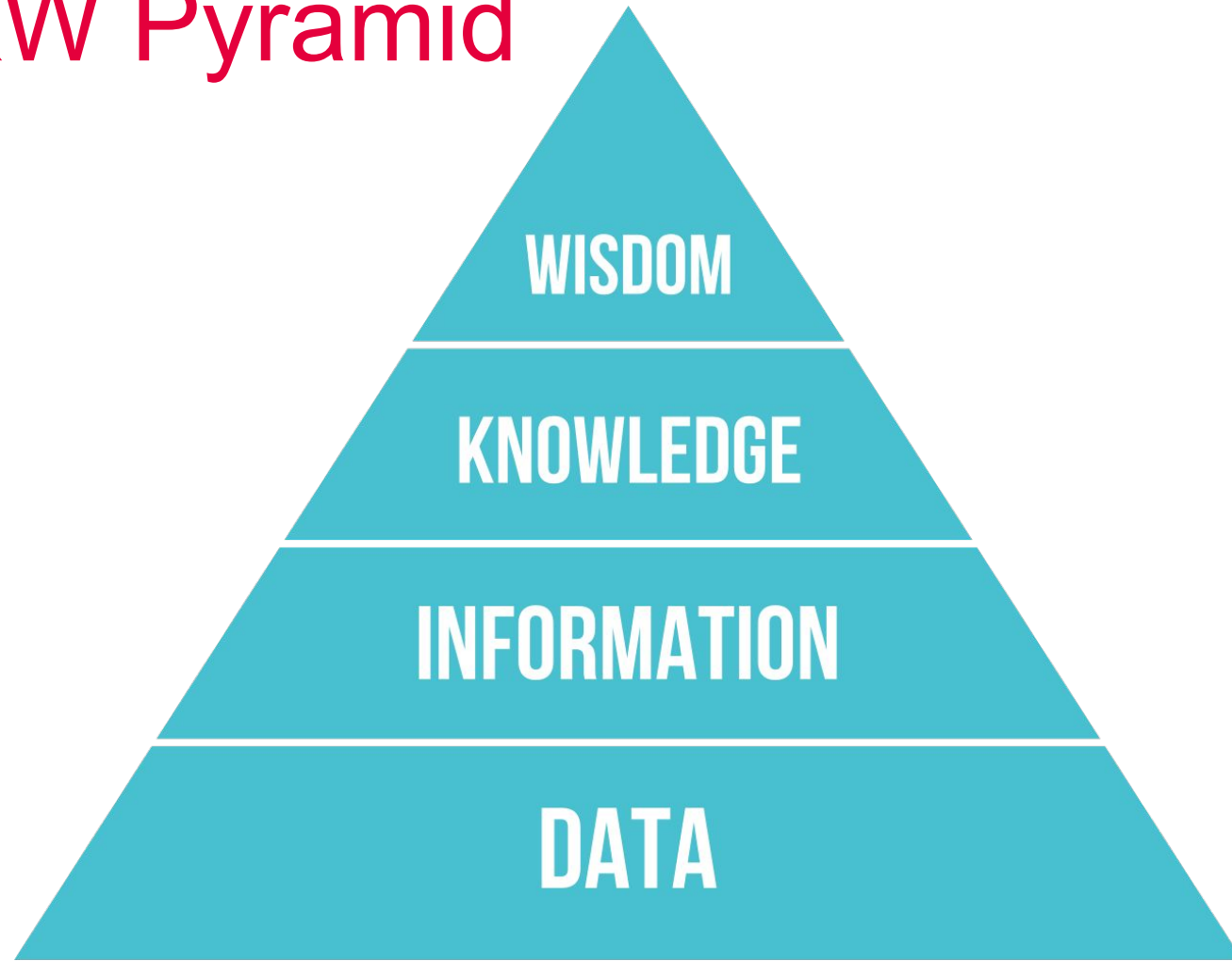
Date: 240 BC

Location: Egypt

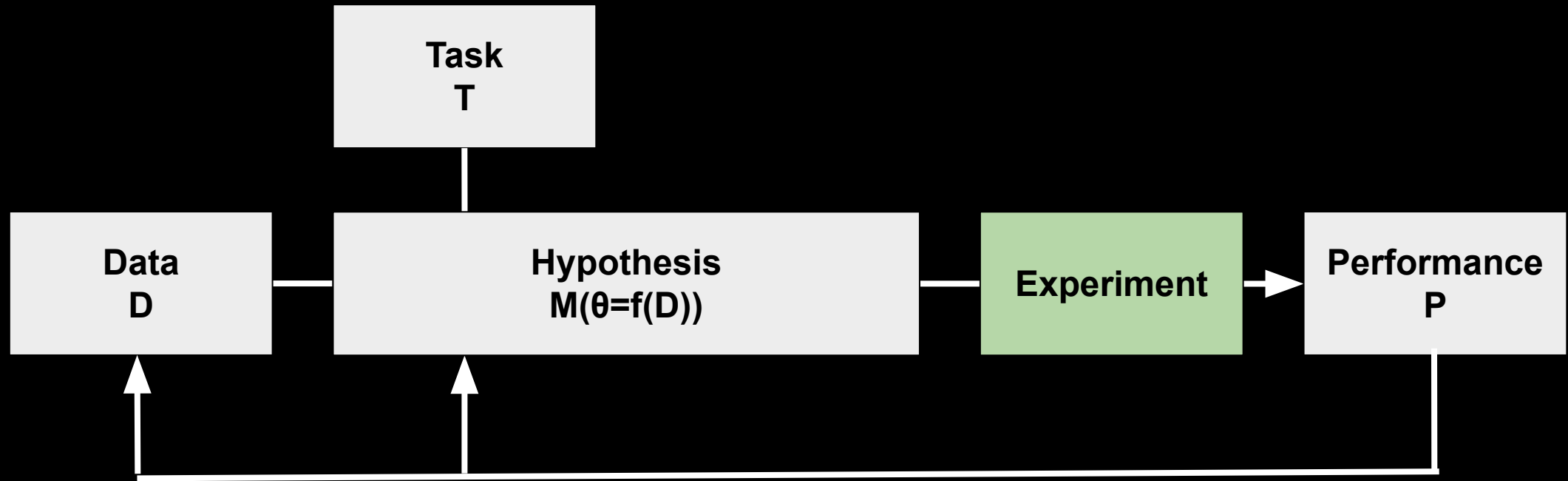
Method: Well, stick, shadow

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The DIKW Pyramid



The Scientific Process / Method



With ML/AI we are increasingly removing the human agent from the process.

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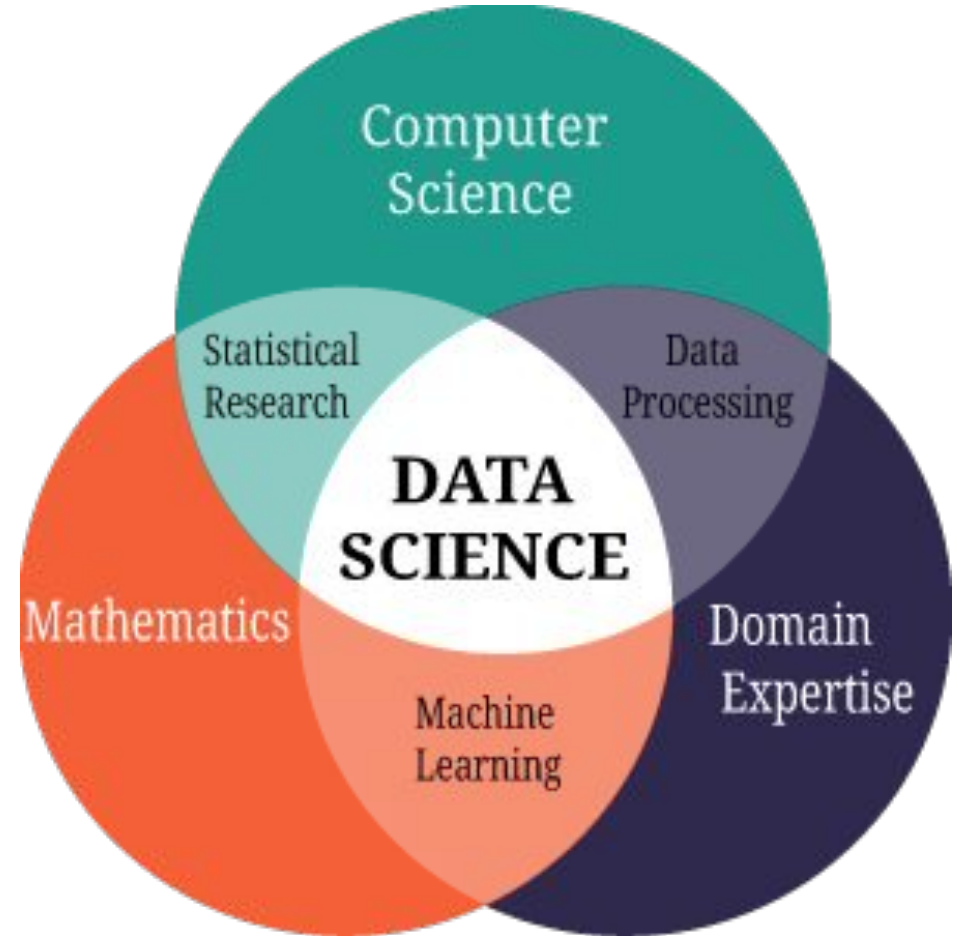
Data Science

The study of the extraction of knowledge from data. It uses

- Mathematics and Statistics
- Computer Science
- Domain expertise

on data to build information and extract knowledge (for decisions and actions).

Very general skills increasingly needed in all empirical research and business.



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Data Quality

Data that enables you to meet your objective. It is “fit” for its purpose. So you have to define the requirements.

Aspects:

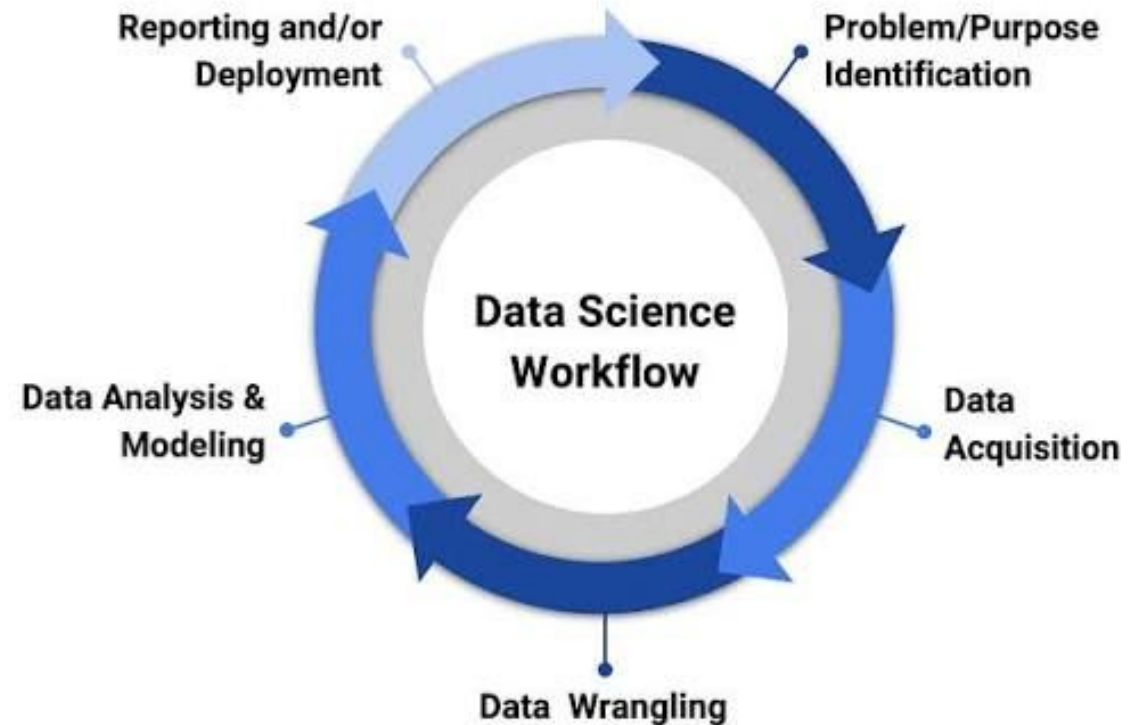
- accessibility or availability
- accuracy or correctness
- comparability
- completeness or comprehensiveness
- consistency, coherence, or clarity
- credibility, reliability, or reputation
- flexibility
- plausibility
- relevance, pertinence, or usefulness
- timeliness or latency
- uniqueness
- validity or reasonableness

In business there is the ISO 8000, less relevant for science.

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Data Science Workflow

- Define objective and requirements
- Data acquisition
- Preprocessing
- Exploration / Descriptive Analysis
- Modelling / Training
- Inference
- Reporting / Deployment



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Summary

- Google Form

Now let's get the hands dirty and start working with the first data and notebook in the GitHub repository.

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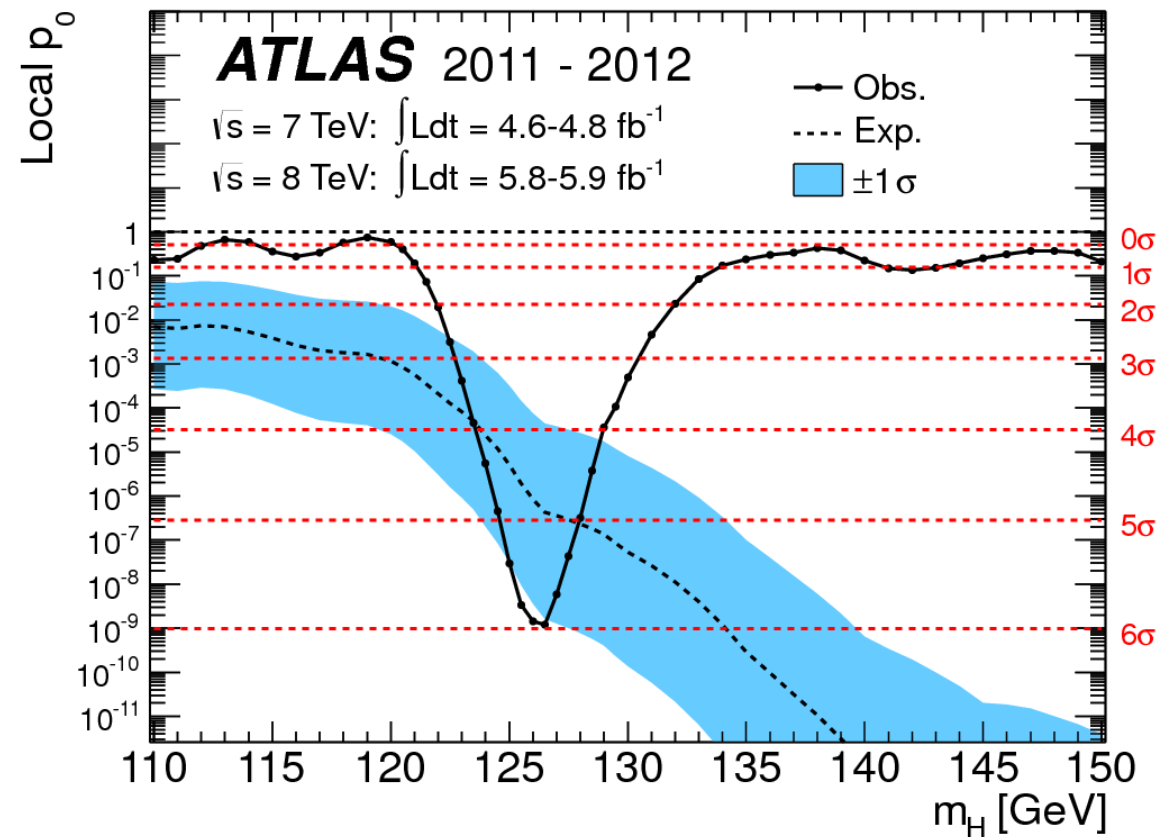
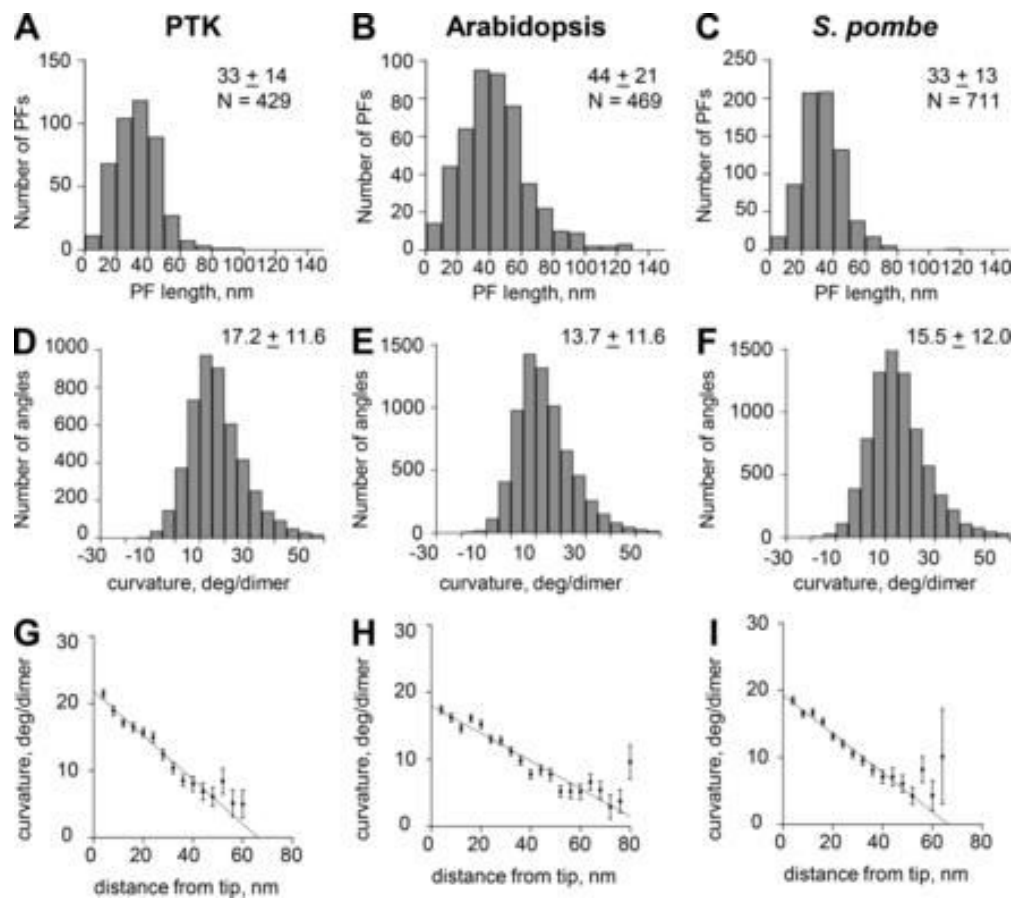
Visualisation of data

Overview [\[edit \]](#)

Data visualization refers to the techniques used to communicate data or information by encoding it as visual objects (e.g., points, lines or bars) contained in graphics. The goal is to communicate information clearly and efficiently to users. It is one of the steps in [data analysis](#) or [data science](#). According to Friedman (2008) the

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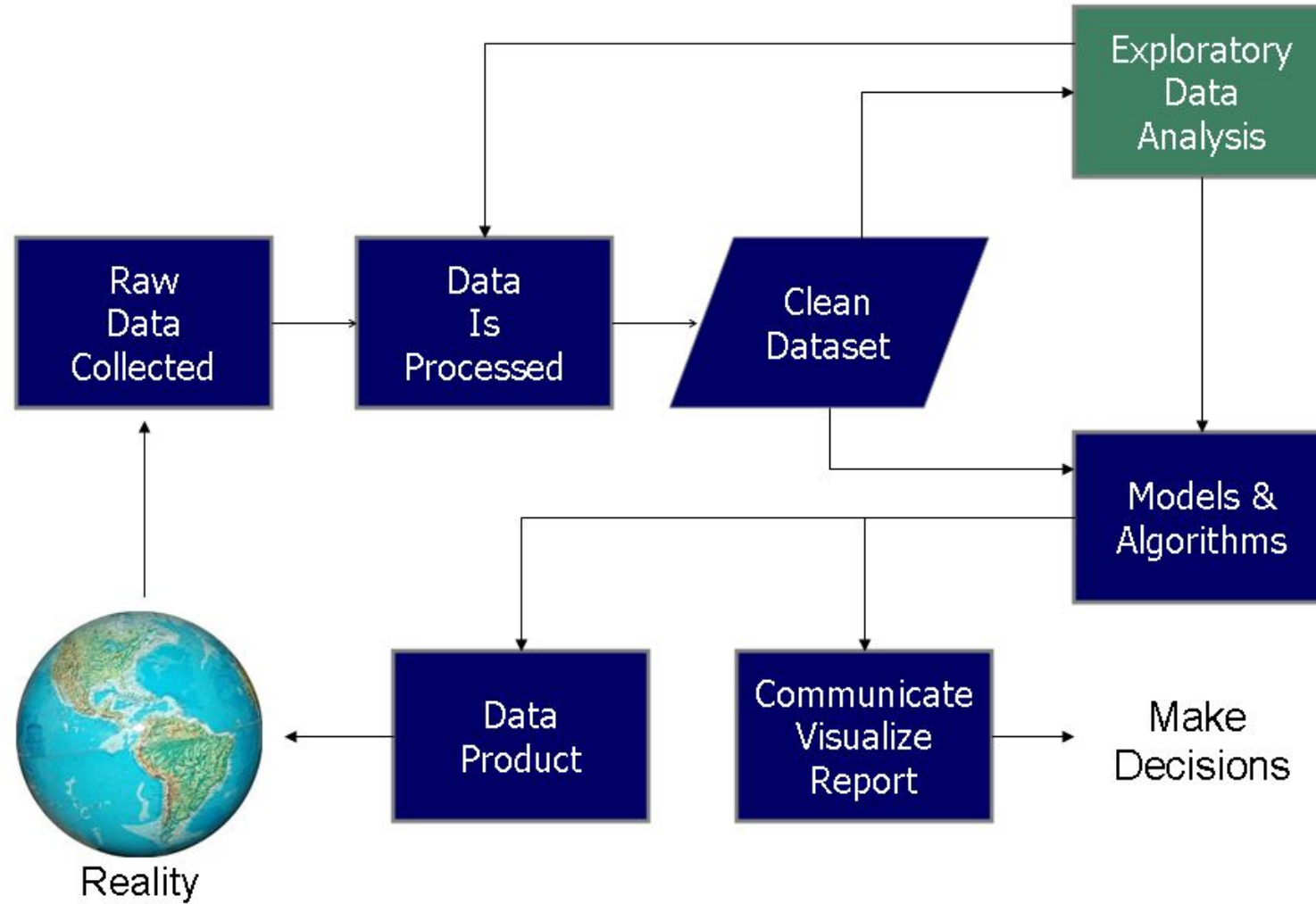
Visualisation examples



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Data Science Process



Example of visualisation in the data science process (actually it is all over)

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Visualisation of data - a division

Descriptive

Describes the data

Helps understand the data

Do as many as possible at every stage

Do different spaces/representations

Look for patterns, similarities
differences, significant features,
correlations ..

Inferential

Communicates information and
knowledge inferred from the data

Can be complex / compact

Normally your “goal” - whole data science
process is about improving the final
inferential graph

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How do you read a publication (paper, book...)?

Typically

Quickly read the abstract

Scan introduction and conclusion (for important numbers)

Study **figures and graphs**

Study tables

Check if there are known references

Dig into the text

So visualisation is important

People with power don't have time

Normally your space and time for communication are limited

Need to pass your message in an elevator (20 seconds)

Good visualisation communicates trust, results and interpretations

Also helps you understand your data

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Visualisation of data - general considerations

Communication

Sender - Message - Receiver

The sender should have a clear motivation and be trustworthy

Choose the right medium for message

The message should be clear and decodable and interpretable for the receiver

Human cognition

Most graphics (still) target humans

Should therefore take

- Cognition
- Pre-attentive attributes

into account

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Visualisation of data - general guidelines

Graphs should reveal data

Show the data

Make the viewer think about the message/data

Avoid distorting from the message/data

Present many number in a small space

Encourage comparison of different pieces of data

Show several levels of detail (from overview to fine structures)

Serve a clear purpose

Be closely integrated with other description of the the data (text, tables etc)

Edward Tufte, The Visual Display of Quantitative Information, 1983

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Visualisation of data - concrete guidelines

Important points for graphs

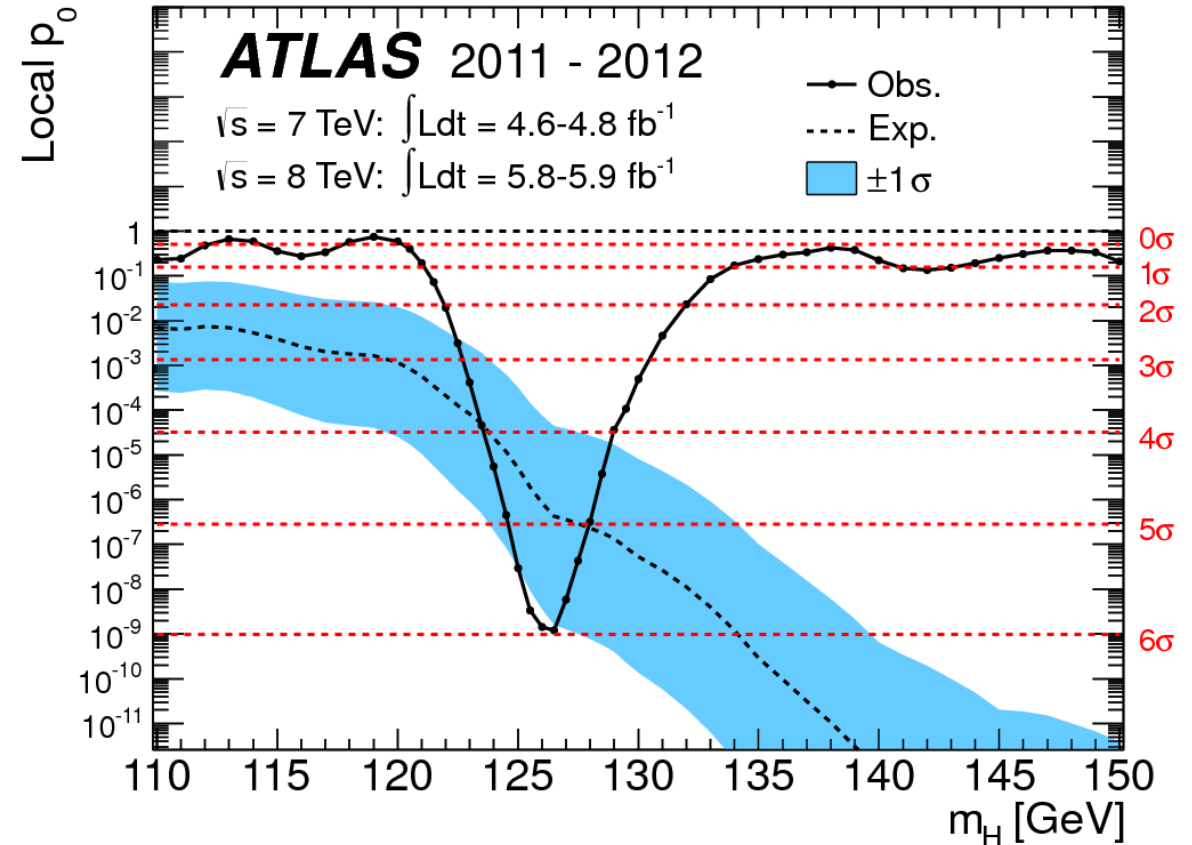
Axes labeling with units

Sufficient but not redundant information for understanding

Readability and visibility

In publications **figure legend** and **reference in text**

In science very often **uncertainties** should be included



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Visualisation of data

8 message types and graphs

Time series

Ranking

Part to whole

Deviation

Frequency distributions

Correlation

Nominal comparison

Geospatial and geographic

[Stephen Few-Perceptual Edge-Selecting the Right Graph for Your Message-2004](#)

For sure there are more ...

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Visualisation tools

Spreadsheets

Easy plotting by mouse clicking

Limited customisation possibilities

In (hard) sciences often below standard

**We don't cover drawings, sketches
etc**

R, Python etc

Programmable plotting

Highly customisable and
automisable

With effort almost everything
can be done
(scripted/programmed)

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Visualisation of data - graphs

Line charts

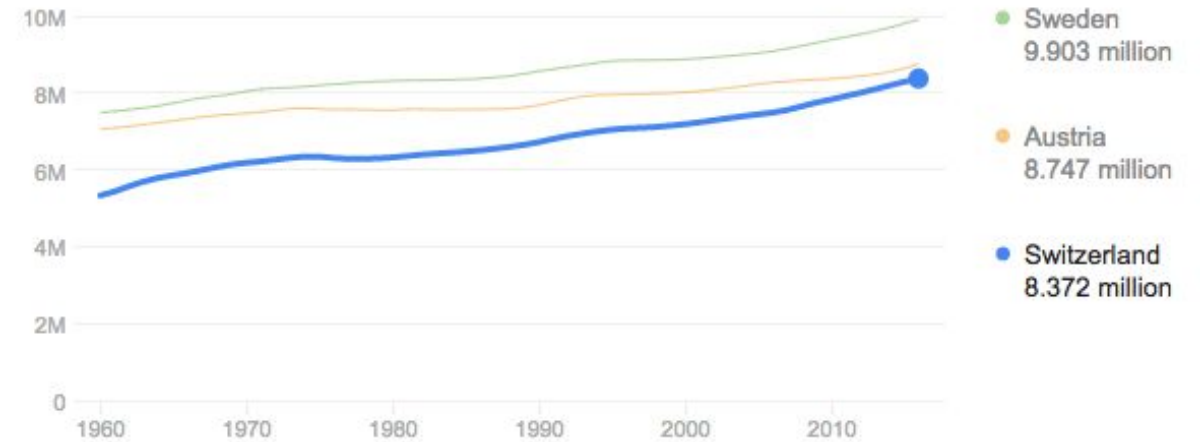
Time series

How a variable varies with time

Example - CH population

Switzerland / Population

8.372 million (2016)



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Graphs - histograms

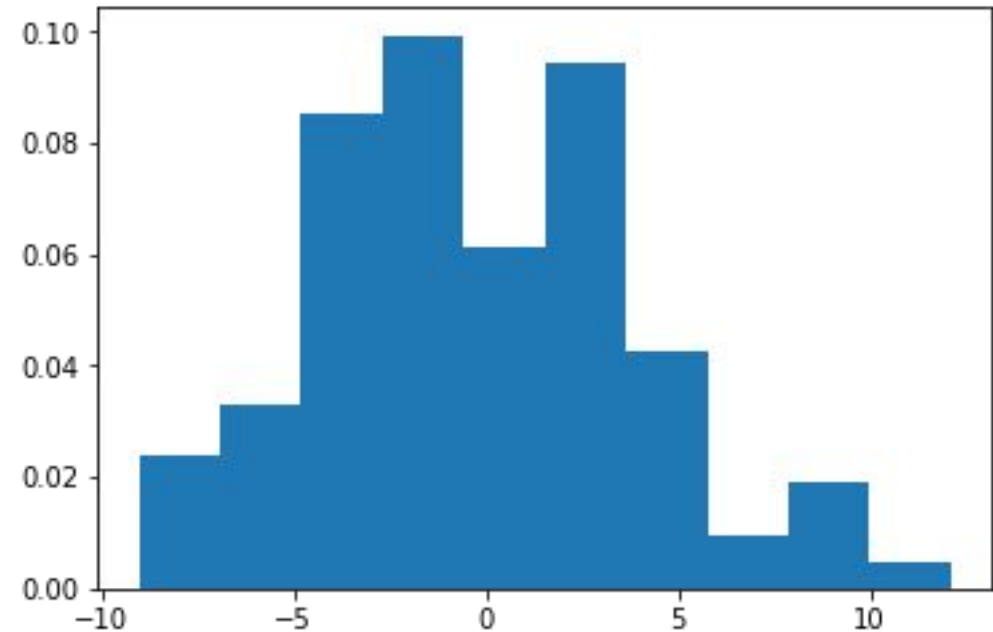
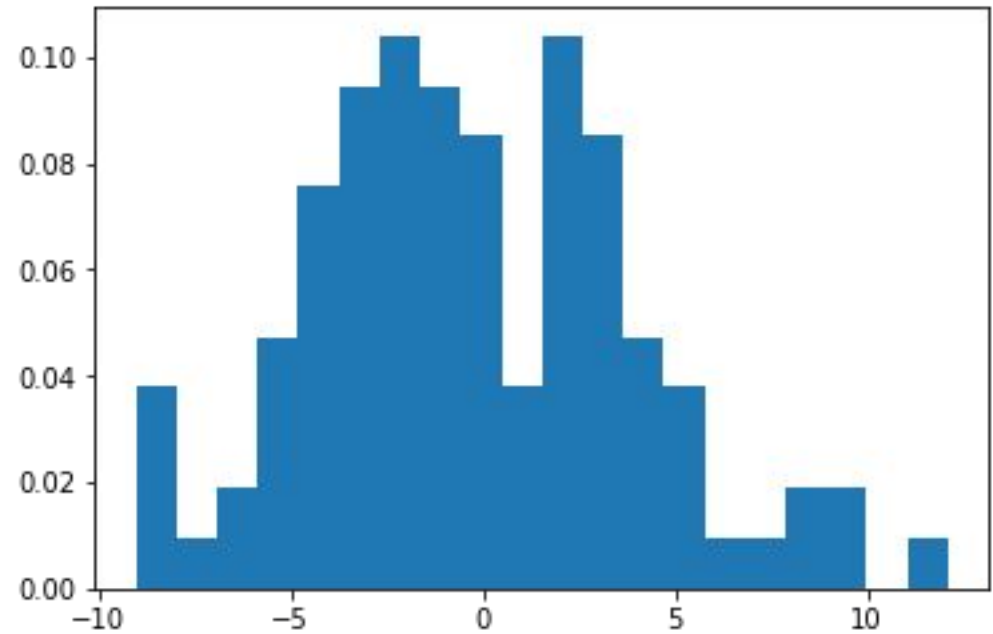
Frequency distribution

Samples the data into bins

Shows the amount of data in each bin

Many bins increase the visual fluctuation

Few bins may hide structures



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Graphs - scatter plots

Shows correlations

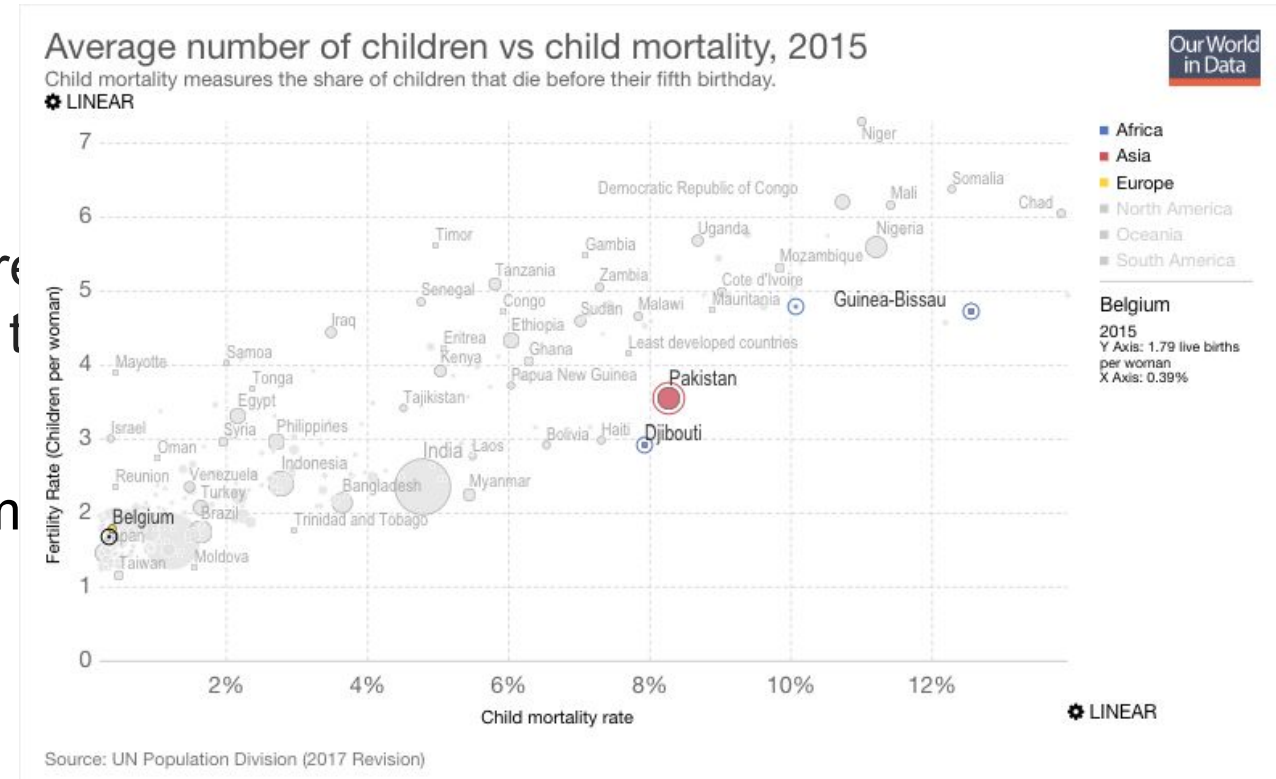
Comparison between observations represented by two variables (X,Y) to determine if they tend to move in the same or opposite directions

Example - human fertility versus child mortality

Scatter plots are often used

Can be 2 or 3 dimensional

Box plots may indicate frequency too



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Graphs - pie charts

Parts-to-whole

Categorical subdivisions are measured as a ratio to the whole (i.e., a percentage out of 100%)

Example - countries with their parts of the total english speaking world population

Pie charts can be used

